



City Plan/ finance stuff\$\$

**Made by Armaan, Partap
and Ammar
(special thanks to Kimayra)**

Intro

Carzanart is a city that will include a city hall where important meetings will take place. The city will mostly be residential. A main part of the city will be the infrastructure that consists of hospitals, schools and other key buildings. We will use unique and intricate designs for our infrastructure such as murals on community centres and adding splashes of color to increase the quality of life. Residential areas will have mixed housing and mixed in with infrastructure and commercial components. Carzanarts biggest company, Saar, produces dinosaur toys, that being the city's biggest supplier. The dinosaur toys are beneficial to the budget and are brought to other countries to strengthen alliance, Carzanart is focusing on bigger and efficient ships to export more goods, and make better income.

Physical Environment

We designed our city to be a region of natural beauty. The city features multi-season tourism attractions providing economic opportunities and work-life balance to the residents. This district includes a small yet historically rich town known as Nevereston. One key natural feature is Mt. Neverest, a mountain located in northwestern Carzanart. Its constant cold temperature allows snow to form. Avalanches carry snow down to a lower, warmer altitude where it gradually melts to form Lake Carzanart. This lake then flows into the St. Car River system, which serves as the city's main water source. The cold and wintry climate of Austria was chosen because it was a middle ground with all the seasons, allowing for year-round productivity, vegetation, and tourism.

To visualize our city, Carznart, we created a 3D model based on a slab of plywood. Our model highlights key features, including residential areas and essential buildings, with labels and clear textures. A government building is included for administrative functions, along with infrastructure for the community's recreation and education. Modeling is an important part of our project, as it showcases the culture and history of Carznart. It consists of a color-printed paper base with a proper and organized street structure and 3D-printed buildings with eye-catching 1:3 ratio sizes (1 cm² base to 3 cm² tall sides). For example, rooms tend to be designed in a ratio of a height of 1x and a width of 3x. Transportation routes will be a mandatory component for cars, buses, and bikes. We promote the use of fewer cars. We also incorporate metro systems and promote cycling, reducing pollution and carbon footprint.

Water Systems/Managing

To protect and contain the water system safely, we will supply ourselves with water from the St. Car River system. As mentioned previously, Mt. Neverest, a mountain in northern Carzanart, contains a waterfall that provides more than enough water for residents and tourists. To tackle the problem of water, we have created the West Carzanart Water Tower to store and preserve our freshwater. To maximize the economy, water exports will be transported along the St. Car river. As a sustainable and smart city, we rely on hydro power to provide electricity renewably. We will keep the river fresh and pure by educating residents on proper waste disposal. We will also reduce pollution by having fines in place. We will also use drainage filters in sewer systems to add another layer of protection to keep the water clean. To prevent erosion in the river, we will add vegetation to bind soil from falling into the river. In order to prevent floods from the river and lake, we will use storm drains and position our residential areas on higher ground.

Climate Regions

Our city is located in Northern Austria, and we experience a cold winter climate along with occasional hot summer heat waves. The climate affects and changes the building style. Transportation will be affected due to the abundance of snow; cyclists won't be able to ride in the cold climate. To adapt to Carzanart's snowy climate, we developed specialized winter bicycle wheels created to grip icy terrain. This allows people to cycle all the way through the city year-round while reducing energy use and pollution from motorized vehicles. We can incorporate excess heat from power plants to heat homes. Despite the limited need for cooling in Carzanart, we will still have the St. Car River system along with transpiration from plants to help regulate the occasional hot summer temperatures.

Vegetation

Despite Carzanart's cold climate, Carzanart makes an effort to incorporate native vegetation throughout the city. One strategy is green roofs with deciduous plants, reducing pollution and providing cooling through transpiration during the summer months. In addition, the city promotes urban farming to increase access to locally produced food and lower the demand from external sources. A large rural area in Southern Carzanart has also been designated for future urban development, which could support either residential or commercial use through urban farming. This reduces burning fossil fuels from long-distance transportation while providing local, fresh food for the community.

On top of the vegetation in an urban setting, Carzanart has a beautiful tourist section, as stated earlier. This tourist destination is an ecosystem consisting of beautiful Mt. Neverest and forest surroundings. We fight for wildlife to have a habitat. We protect by preventing deforestation and soil erosion, and do so by having check-ups from the World Wide Fund for Wildlife(WWF), Green Belt, and the Region of Carzanart to leave old trees untouched.

Human Impact & Sustainability

Carzanart faces environmental difficulties year-round due to the cold and snowy climate including frozen rivers, avalanches, and flooding. Our city reduces and minimizes these problems by keeping around heating systems and manufacturing more coats for people going outside. These coats will be made out of sealed biomass, a biodegradable material under conditions. To fight the avalanches, dams were installed and avalanche tests were run on the main parts of the mountain. We proceed with these strategies to identify and reduce the amount of avalanches or damage. For residential areas near Mt. Neverest, we designed bunkers to survive avalanches.

For us to reduce the amount of fossil fuels used, we recommend bikes to people or use public transit to bring down air pollution and global warming. We will also counter air pollution with plants to absorb CO₂. We incorporated visible signs that the forest had been reduced to the north due to human impact; it previously covered the area of the city and was reduced to create houses, like in modern-day cities. To sustainably get energy, we choose to use renewable energy by utilizing solar and hydro sources. To source solar energy, we utilize the high altitude of Mt. Neverest and the highly reflective, white snow.

Reflection

Why is it important to plan cities sustainably?

It is important to design cities sustainably so that tax payer money is spent efficiently, cities can manage dense populations, and power isn't limited. We plan it sustainably by having cities planned with roads and intersections to be placed in proper areas to prevent urban sprawl(eg. India). To manage dense populations in Carzanart we incorporate mixed housing like condos, houses, and hotels for tourists so that there is affordability and quality for residents. In order to have long lasting energy, we incorporate renewable energy sources of hydroelectric sources.

How does geography help people make better decisions?

We learned that geography helps people make better decisions by understanding how the slightest miscalculations can lead to urban sprawl. We also learned that not only precision was key but building design was also a huge factor. Including white buildings with green roofs made our model a more eye-catching model in comparison to yellow buildings.

What was the biggest challenge in designing your city?

The biggest challenge in designing our city was proportionality and road structure. We tackled this issue by virtually designing our city on a Windows app known as Inkscape and color printing slices out at Staples. In order to fix proportionality we measured the board and added 3D prints according to the proportionality ratio of 1:2250.

Bonus: What did we learn about design and proportionality? Why is it important?

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Residency

- House price:
€1,300,000 Total
- buying fees:
€91,000–€182,000
- Final cost: €1.39M–
€1.48M
- Yearly tax: €10,000–
€15,000



- Base price:
€500,000,000
- Total taxes + fees:
€51,500,000
- Final cost:
€551,500,000
- Yearly cost: €3.75M

Residency

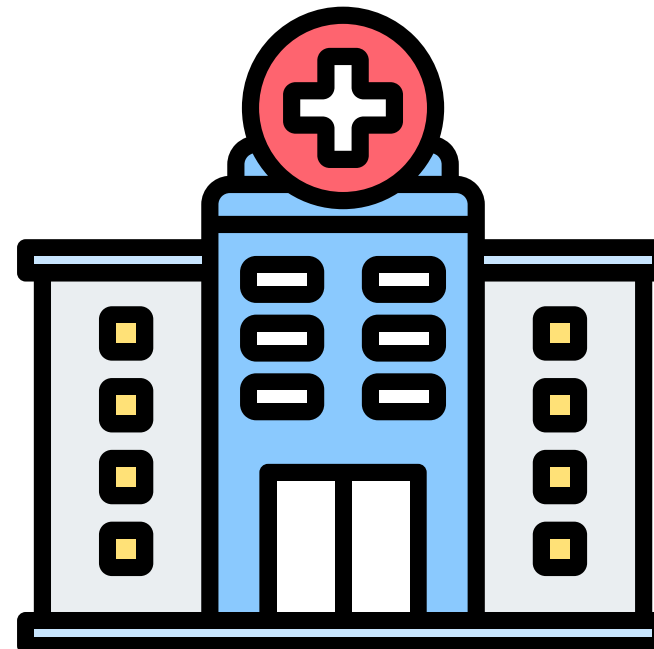
- Build cost: €230,000
- Yearly cost: €30,000



- Base construction: €150M – €300M
- Medical equipment: €80M – €150M
- Taxes + permits + planning: €20M – €40M
- Yearly operating cost: €80M – €150M
- Workers: 800 – 1,500
- €250,000,000 – €490,000,00

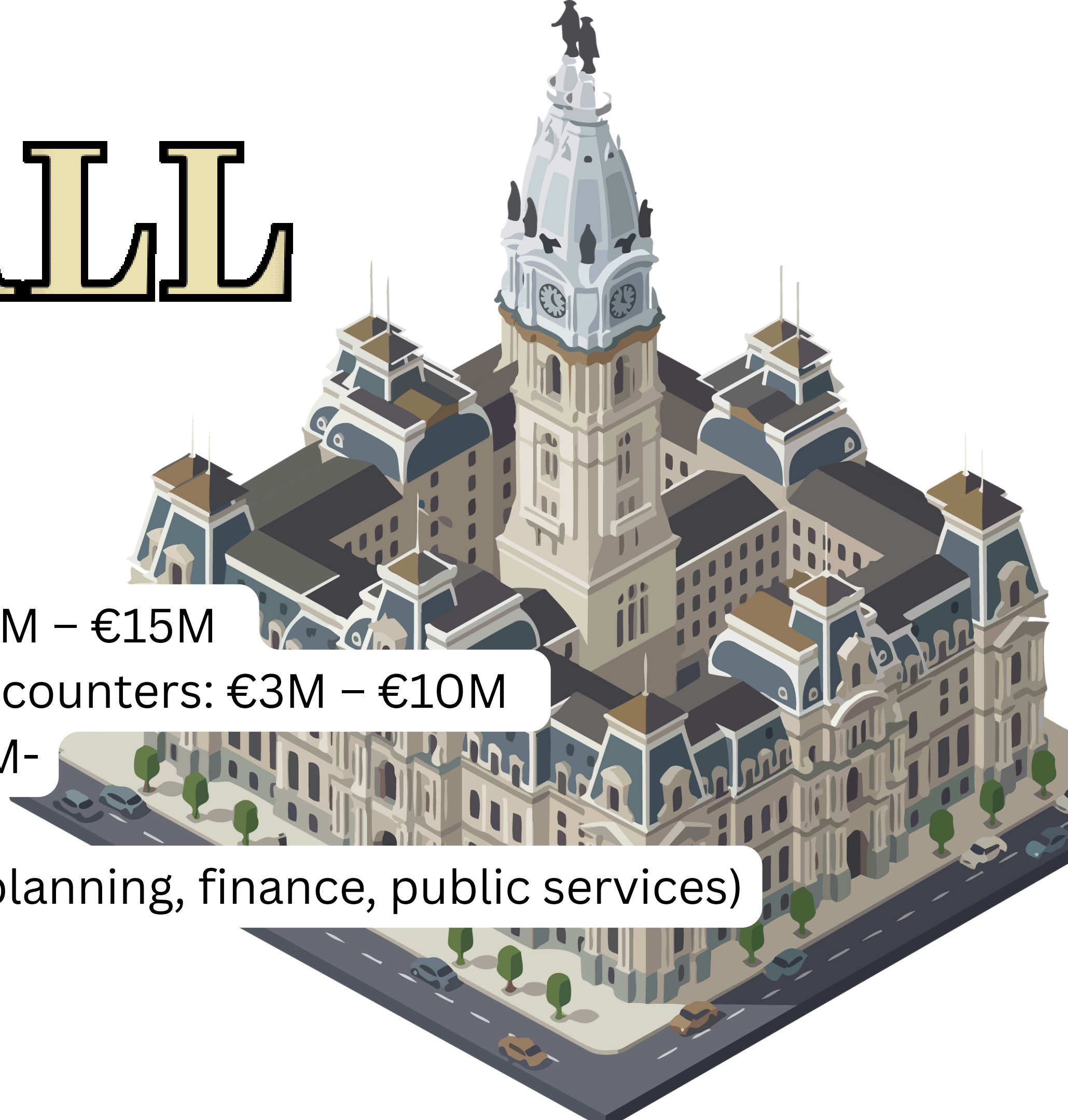


- Base construction: €8M – €15M
- Taxes + permits + planning: €800k – €2M
- FINAL TOTAL COST: €9M – €17M•
- Average final cost: €13M



CITY HALL

- Base construction: €30M – €60M
- Interior + offices + meeting rooms: €5M – €15M
- IT systems + security + public service counters: €3M – €10M
- Taxes + permits + planning: €2M – €5M-
- Yearly operating cost: €5M – €12M
- Workers: 150 – 400 (administration, planning, finance, public services)





Concept/Prototype



final/flag

Coat of arms final





**Thank you for watching this
presentation**

From *Carzanart*